RESEARCH ARTICLE SUMMARY

MICROBIOTA

Imbalance in gut microbial interactions as a marker of health and disease

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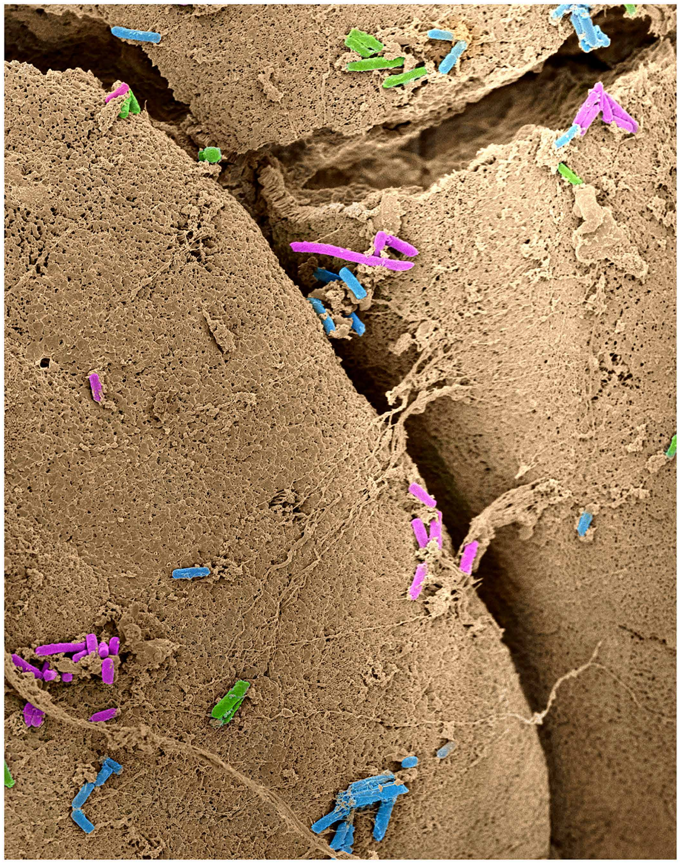
INTRODUCTION: The human gut microbiome is a complex ecological system crucial for host health. Dysbiosis—i.e., the imbalance of gut microbial communities—is associated with a wide range of diseases, including obesity, diabetes, inflammatory bowel disease (IBD), *Clostridioides difficile* infection (CDI), irritable bowel syndrome (IBS), and colorectal cancer (CRC). Therapeutic approaches, such as fecal microbiota transplantation, dietary interventions, and probiotics, aim to restore balance by reshaping community composition. However, their outcomes remain inconsistent and unpredictable, in part owing to our limited understanding of the metabolic and ecological interactions that govern microbiome dynamics. The latter has also prevented the development of robust biomarkers to distinguish health from disease.

RATIONALE: Previous studies have suggested that health and dysbiosis may represent alternative community states but have not provided a mechanistic justification. Most efforts to define dysbiosis aim to identify bacterial taxa or functions that may differ between healthy and diseased communities or assume that reduced diversity is a universal hallmark of disease. However, such signatures vary across conditions and cohorts and fail to capture the ecological principles that shape disease states. To provide a more mechanistic understanding of gut microbial dynamics in health and disease, we developed a metabolically explicit model in which bacterial interactions arise naturally from competition for shared resources and cross-feeding.

RESULTS: The model reproduces key macroecological patterns and captures the functional redundancy characteristic of real gut microbiomes. Moreover, our model revealed the emergence of two distinct ecological states (healthy and dysbiotic states) whose α and β diversities, dominance indices, and numbers of functions and excreted metabolites closely resembled those observed in real microbiomes. The healthy state was dominated by competitive interactions, whereas the dysbiotic state was shaped by tightly connected cross-feeding consortia. We also developed the ecological network balance index (ENBI), a metric that measures the relative contribution of positive versus negative interactions and reliably separates healthy from dysbiotic states. Calculating the ENBI for the model and metagenomic data for IBD, IBS, CDI, and CRC showed that, in all cases, diseased microbiomes exhibited higher ENBI values. Our metric also correlated with disease stage. These results proved robust across subsampling, geography, taxonomic levels, and profiling methods.

CONCLUSION: Unlike diversity-based metrics, which vary across diseases and cohorts, the ENBI consistently distinguishes healthy from diseased states and even tracks disease progression, offering a path toward robust, noninvasive early warning indicators of disease. The ENBI also provides mechanistic insights: Our results show that dysbiosis is associated with a shift in the community interaction network, with positive interactions increasingly dominating over negative ones. Our framework is both general and extensible, and thus it can be adapted to incorporate additional biological features for the study of specific gut phenomena and be readily applied to other microbiomes (from the vaginal and oral to plant and soil ecosystems), or it can be used for the study and prediction of potential outcomes in therapeutic interventions. By linking microbial ecology with clinical research, our framework advances precision medicine and supports the development of more personalized strategies to maintain or restore gut health. □



Consortia of gut bacterial cells on villus tips (projections from intestinal mucosa). Gut bacterial cells are shown in assorted colors, and the villus tips are brown. The color highlights the contrasting size of the bacteria and the barren-appearing cliffs of intestinal cells and chasms. Our model and metric reveal that positive interactions dominate in diseased gut microbiomes, whereas negative interactions dominate in healthy ones. We track disease emergence and progression monitoring this ecological balance. [Image credit: X. Zhang; edits: A. J. Roldán]

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MICROBIOTA

Imbalance in gut microbial interactions as a marker of health and disease

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Imbalances in the human gut microbiome, or dysbioses, are associated with multiple diseases but remain poorly understood. Existing biomarkers of dysbiosis fail to capture the ecological mechanisms that differentiate healthy from diseased microbiomes. We have developed a metric, the ecological network balance index (ENBI), that quantifies the balance between positive and negative microbial interactions. This metric was inspired by the phenomenology observed in a model for gut microbiome dynamics that we introduce in this work, which revealed alternative stable states with distinct emergent microbial communities: a healthy state dominated by negative interactions and a dysbiotic state dominated by positive interactions. The ENBI robustly differentiates these states in both simulated and empirical datasets spanning multiple diseases and correlates with disease progression in conditions such as colorectal cancer, which underscores its potential as a diagnostic tool.

The human gut microbiome is a dynamic and complex system, comprising assemblages of diverse microorganisms, that plays a crucial role in maintaining health through its interactions with the host and environment (*1*–*4*). Innovations in DNA sequencing techniques over the past two decades have improved our ability to quantify microbiome composition and functional capabilities (*5*–*7*). These studies have shown that imbalances in the gut microbiome, or dysbioses, associate with several diseases, including obesity, diabetes, inflammatory bowel disease (IBD), *Clostridioides difficile* infection (CDI), irritable bowel syndrome (IBS), colorectal cancer (CRC), liver disease, pancreatic disorders, psoriatic arthritis, and celiac disease (*1*, *2*, *5*, *8*–*13*). Treatments, such as fecal microbiota transplantation (FMT), diet-based interventions, and probiotics (*14*–*16*), aim to restore microbial balance and reverse dysbiosis by altering community composition; however, these treatments show highly variable and unpredictable success rates. One reason for this unpredictability is the limited understanding of the mechanisms underlying gut microbiome dynamics and structure. Specifically, the metabolic and ecological interactions among microorganisms are poorly understood (*1*, *17*, *18*). This knowledge gap has hindered the identification of microbiome biomarkers that correlate with—and accurately differentiate—health and disease. Our work seeks to address both of these fundamental challenges.

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The search for a universal indicator of dysbiosis

Dysbiosis is a broad concept that encompasses various compositional and functional attributes across diseases, and it is not unambiguously characterized (*10*). Early efforts focused on identifying single bacterial species or functions whose altered levels correlate with a particular disease (*8*, *10*, *19*, *20*). However, dysbiosis typically affects multiple species and functions and can only be captured by systemic, whole-community approaches (*13*, *21*, *22*). A common claim across studies and conditions is that disease correlates with reduced microbial diversity, which is often suggested as a universal marker of dysbiosis (*5*, *11*, *12*, *23*–*25*). We reexamined empirical data from several diseases and found a more nuanced pattern (Fig. 1), in line with recent studies (*10*, *13*, *21*, *26*). Specifically, although IBD and CDI (Fig. 1, A and B) consistently show reduced microbial diversity compared with healthy microbiomes (*13*, *27*, *28*), conditions such as IBS and CRC (Fig. 1, C and D) show a range of microbial diversity and were less clear-cut (*13*, *29*, *30*). Therefore, the question remains as to whether there are conserved mechanisms driving dysbiosis, which would enable the identification of robust disease indicators.

The dichotomy between health and disease has fueled the hypothesis that dysbiosis emerges from shifts between alternative stable states—that is, between distinct taxonomic or functional configurations of the gut microbiome (*31*–*37*). However, this idea remains under debate owing to the varied methodologies used to define and measure such states; the context-dependent nature of the microbiome; and the challenges of controlling multiple external factors, such as diet, medication, and lifestyle (*5*, *8*, *38*). All these factors influence gut composition and function, which complicates the identification of stable states and consistent disease markers.

Theoretical models, nonetheless, allow us to fine-tune definitions and environmental factors (*18*, *38*–*41*). For example, models have identified microbial taxa capable of inhibiting *C. difficile* (CD), a prediction subsequently validated experimentally to confirm that the identified taxa indeed conferred CD colonization resistance (*42*). In another example, machine learning models have systematically tested competing hypotheses regarding the success of FMT (*43*). Despite these advances, few models designed to study gut microbiome dynamics predict or generate alternative stable states, which limits our understanding of their emergence and fundamental properties (*39*, *44*). To address this gap, we introduce a mathematical model specifically designed to explore the ecological dynamics of the gut microbiome and mechanisms underlying its emerging collective states [Fig. 2A and materials and methods (*45*)].

A model for the dynamics of the gut microbiome

Our consumer-resource model describes bacterial and nutrient dynamics and incorporates the possibility of cross-feeding as new metabolites are produced by different bacterial strains (see materials and methods). Nutrients are introduced at a rate h (representing host diet); new bacterial species immigrate at a rate U; and bacteria, nutrients, and metabolites are removed at a dilution rate δ, which determines intestinal transit time.

We also aim to capture the role that metabolic pathways play in shaping gut communities (*46*–*50*). Thus, in our model, each bacterial lineage is characterized by the set of metabolic transformations that it can perform, defined by the network of possible conversions between nutrients and metabolites, along with the metabolic preferences specific to the bacterial strain. Although, for simplicity, only catabolic reactions are explicitly modeled, we use a trade-off function to assign a cost to the set of available pathways; this cost arises from the energy allocated to constructive metabolism and therefore effectively implements anabolism. Thus, our model captures both the energy gain and enzymatic cost for each metabolic reaction.

To model the metabolic conversion of a substance S_α into another substance S_β, we monitor (i) the free energy ($E_{\alpha\beta} > 0$) released during

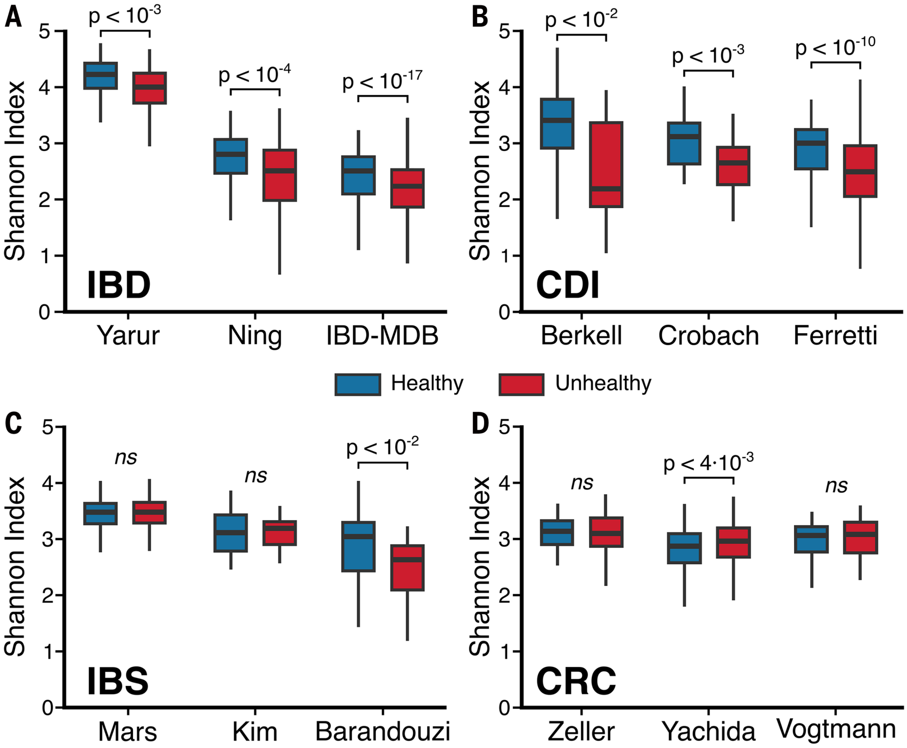

Fig. 1. α diversity in healthy and unhealthy individuals. (**A** to **D**) Comparison of the Shannon index [see materials and methods (*45*)] obtained using existing metagenomic analyses from healthy individuals (blue) versus individuals affected by a specific disease (red): (A) IBD, using data from Yarur *et al*., Ning *et al*., and the Human Microbiome Project (*84*–*86*). (B) CDI, using data from Berkell *et al*., Crobach *et al*., and Ferretti *et al*. (*87*–*89*). (C) IBS, using data from Mars *et al*., Kim *et al*., and Barandouzi *et al*. (*90*–*92*). (D) CRC, using data from Zeller *et al*., Yachida *et al*., and Vogtmann *et al*. (*30*, *93*–*95*). A significant decrease in diversity is evident for IBD and CDI but not for IBS or CRC (statistical significance calculated with two-sided Mann-Whitney *U* test; ns, not significant).

the process and (ii) the enzymatic cost ($C_{\alpha\beta} > 0$), reflecting the number and complexity of enzymes involved. Each bacterial lineage i is defined by a unique pathway matrix $P^i_{\alpha\beta}$ (random binary matrix with the metabolic routes available to that bacterial lineage) and a coefficient γ_i that controls the fraction of free energy used for biomass growth. Finally, we used a bottom-up approach to parameterize the model, by which we assigned biologically plausible values to the parameters instead of relying on data fitting exercises. A detailed description of the model is included in the materials and methods, including the definition of species, nutrients, metabolites, the implementation of bacteria-specific metabolic preferences, species immigration, pathways, trade-off function, and chosen parameterization.

Our model successfully reproduced well-known macroecological patterns of the gut microbiome that are observed empirically across multiple systems. In particular, the 95% confidence interval obtained across replicates of the model captures the distributions of species abundance mean and fluctuations, Taylor's law (i.e., the relationship between mean and variance), species prevalence (the distribution of the fraction of samples in which a species is present), and the dependence of prevalence on mean abundance (fig. S1) (*51*, *52*). Additionally, fig. S2 shows that our model replicates the functional redundancy observed in the gut microbiome, where multiple taxonomic compositions encode the same metabolic functions (*53*). In our framework, functions are represented by metabolites grouped into coarse-grained energy levels (*45*). These broader functional categories are akin to Clusters of Orthologous Groups of proteins (COG) or Kyoto Encyclopedia of Genes and Genomes (KEGG) classifications for empirical data (*54*, *55*). Functional redundancy becomes apparent at sufficiently coarse-grained resolutions for both our model and observed data.

Alternative states in the gut microbiome model may reflect health and disease

The dynamics of our simulated gut microbiome community suggest two qualitatively distinct classes of states (Fig. 2B): one characterized by the rapid turnover of many bacterial lineages and another where only a few lineages dominate with quasi-stationary abundances (*45*). We labeled these two classes of states as healthy and dysbiotic, respectively, for reasons that will become apparent in comparison with real data. Notably, these alternative states emerge consistently across a wide range of parameter values, which indicates that they result from the mechanisms encoded in the model rather than specific parameterizations (figs. S3 to S6).

Existing community-level theoretical models, such as the generalized Lotka-Volterra (gLV) (*56*, *57*) and consumer-resource models with random interactions (*58*, *59*), report dynamics reminiscent of either one of our two states (often referred to as "chaotic fluctuations" and "stable fixed points"). However, these models do not capture multistability between the different states. Explicitly incorporating metabolic pathways and letting the structure of the community emerge from the dynamic interactions allowed our model to capture diverse alternative states and the dynamic transitions between them.

Compared with the healthy state, the dysbiotic state in our model exhibited a significant reduction in α diversity (measured by the Shannon index; richness or number of lineages; Simpson's index; and Pileou's evenness) (Fig. 2, C and D, and fig. S7A), the number of enzymes (total enzymatic cost of all active pathways), the number of secreted metabolites (substances with nonzero concentrations), and an increase in dominance (relative abundance of most abundant bacteria) (*45*). These findings indicated that the microbial community in the dysbiotic state operates with greater efficiency compared with the community in the healthy state because fewer bacterial lineages and metabolic pathways are required to form consortia able to convert a larger proportion of the available energy into microbial biomass (fig. S8).

Given the dichotomy between these two distinct states, we also measured these biomarkers in healthy individuals and patients with IBD. The resulting pattern mirrors our model predictions (Fig. 2E and fig. S7B), supporting our labeling of the two alternative states of the model as healthy and dysbiotic. The parallelism of model states with real healthy and dysbiotic states is reinforced by the observation that in the model and in IBD, the shift toward dysbiosis involves the loss of specific metabolic pathways, resulting in a disruption of functional capacity (that is, some functions may be impaired or entirely lost, whereas others become overrepresented; Fig. 2, F and G) (*60*). Furthermore, for our model, as well as for a range of diseases, β diversity across healthy states was significantly lower than for dysbiotic states (fig. S9), which indicates that healthy microbiomes are more similar to each other than the similarity shown by dysbiotic communities.

The patterns mentioned above, however, are subtle or inconsistent across diseases (see, for example, β diversity for CRC in fig. S12D or the number of enzymes, diversity of secreted metabolites, richness, dominance, Simpson's index, and evenness for IBS and CRC in figs. S10

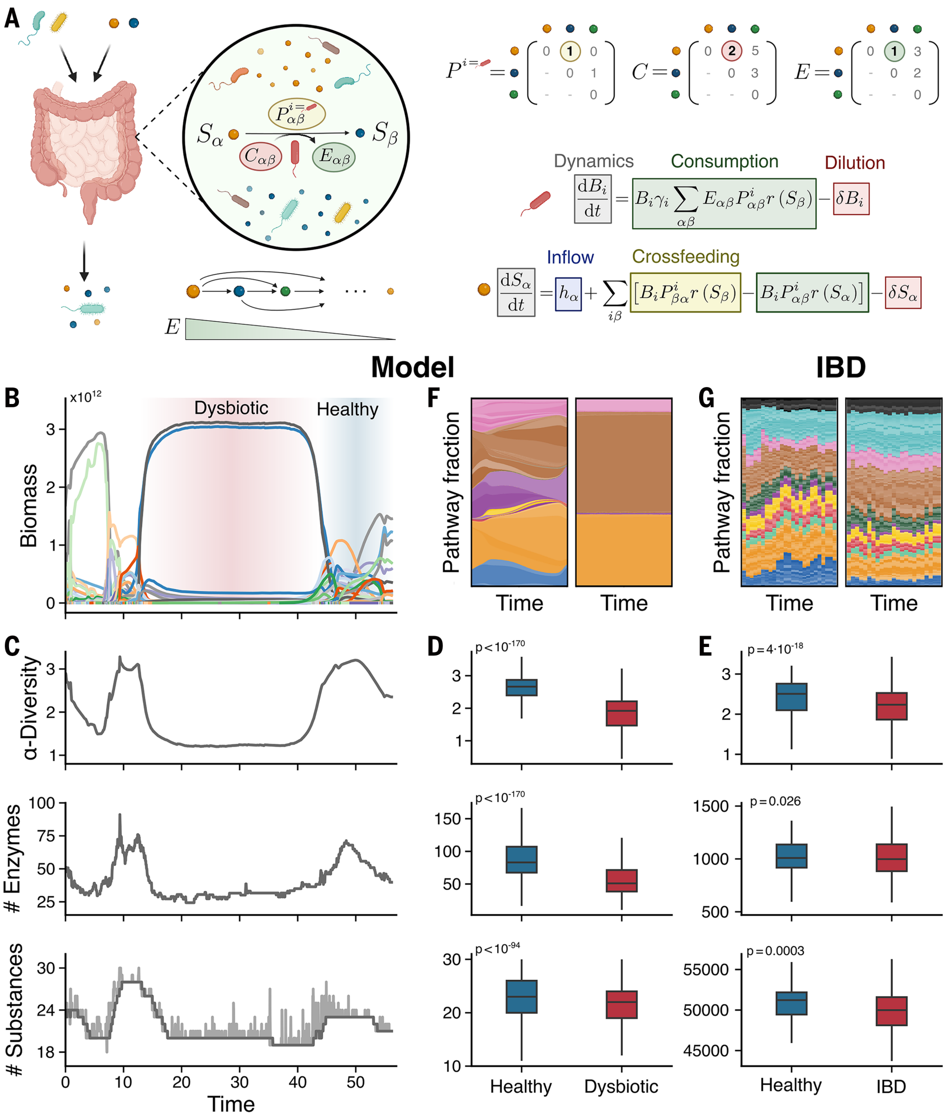

Fig. 2. Gut microbiome model and the spontaneous emergence of alternative stable states. (**A**) Schematic representation of the model. The gut microbiome is conceptualized as living in a chemostat-like environment that includes both bacterial species and substances. Metabolic pathways linking any two substances, S_α and S_β, are characterized by their enzymatic cost, $C_{\alpha\beta}$, and the energy obtained from the conversion, $E_{\alpha\beta}$. Each bacterial species can use specific pathways depending on its pathway matrix, P^i, which determines the metabolic routes available to that species. The model considers only catabolic pathways, where $E_{\alpha\beta} > 0$. (**B**) Biomass (cells) over time for a realization of the model, illustrating the emergence of dysbiotic and healthy states. (**C**) For the model simulation in (B), α diversity (top), number of enzymes (middle), and number of substances (bottom) over time. Dark gray represents rolling average, and light gray represents the actual values. (**D**) Comparison, for the same metrics in (C), of healthy and dysbiotic states in the model. (**E**) Corresponding metrics using existing data for healthy individuals and IBD patients. (**F**) Coarse-grained pathway fraction over time for a healthy (left) and dysbiotic (right) state in the model; colors represent different (coarse-grained) functions. (**G**) Similar plot for healthy versus IBD patients, where colors represent (coarse-grained) functions, and shades represent the various pathways belonging to the given function. All P values show two-sided Mann-Whitney U tests. Further details are provided in the materials and methods (*45*).

and S7). Beyond this lack of consistency, these metrics merely reflect the shift from one state to the other and do not explain the mechanisms underlying the transitions, nor can they be used as early warning indicators.

Thus, in the search for a more mechanistic characterization of these shifts in the model, we analyzed the pathway structure for healthy and dysbiotic states (fig. S11). We observed that dysbiotic states contained self-sufficient consortia with minimal pathway overlap—i.e., consisting of bacterial lineages that metabolized the most energetic resources through fewer, more direct metabolic routes, thus enabling efficient exploitation of available resources (fig. S11, A and B) and potentially facilitating host exploitation (*61*). Bacteria in healthy states, by contrast, showed greater functional redundancy, with overlapping metabolic pathways. Bacterial lineages in dysbiotic states were typically closer to the theoretical optimum of the enzymatic trade-off function compared with lineages in healthy states (fig. S11C), which explains the increased persistence (fig. S12A) and dominance (fig. S7A) observed in dysbiosis. Examples of bacteria for which an increased presence has been reported to correlate with real diseases include *Bacteroides fragilis* or *Escherichia coli* in IBD and *Fusobacterium nucleatum* and *Solobacterium moorei* in CRC (*30*, *62*).

Moreover, the model offers mechanistic insight into how transitions between states occur. Transitions to dysbiosis are typically triggered by metabolically compatible bacterial consortia that form a tightly connected metabolic loop or short chain, efficiently metabolizing resources with minimal pathway overlap (fig. S13) and allowing them to rapidly dominate the community. By contrast, recovery from a dysbiotic state requires the establishment of bacterial taxa that can disrupt the metabolic loop or chain of the dominant consortium and establish new pathway interdependence among themselves, thus leading to a collective reorganization of the community structure. These dynamics suggest a possible mechanism for the existence of disease-associated taxa observed across experiments (*10*, *21*) and may help explain why FMT or multistrain probiotic interventions can be effective (*14*, *16*).

Community interactions are disrupted in dysbiotic states and disease

Microbial communities are structured not only by environmental conditions but also by ecological interactions, including competition, cooperation, and cross-feeding (*63*, *64*). Experiments with synthetic communities show that competition often dominates, yet cooperative behaviors such as cross-feeding can arise under certain constraints and promote coexistence (*65*–*67*). Natural communities typically show both interaction types, and their dominance in the community dynamics may shift depending on environmental context (*68*, *69*). Beyond pairwise interactions, recent studies have underscored the role of higher-order effects and ecological feedbacks in community dynamics (*70*, *71*). Together, these findings underscore the importance of explicitly modeling ecological interactions to understand the emergent behaviors of microbial ecosystems.

In our model, bacterial interactions emerge from competition for shared resources and cross-feeding. Community dynamics ultimately determine which substances (and, indirectly, which interactions) persist in the system. Therefore, our model enables a shift of focus from broad measures of species and functional diversity to the emergent, dynamic ecological network of effective interactions that collectively shape the resulting community (*39*, *72*).

There is currently no consensus on whether competition or cooperation dominates gut microbiota dynamics. Although some empirical studies suggest that competitive interactions dominate owing to their role in promoting stability (*18*, *63*, *73*), others emphasize the critical role of cross-feeding interactions in maintaining a healthy microbial community (*74*–*77*). The relative contributions of these interactions to gut microbiota function in health and disease remain unclear.

To quantify contributions of competition and cross-feeding, we used a net interaction metric ρ, defined as the difference between the total cross-feeding and competitive interactions within the community, normalized by the sum of all interaction strengths [see eq. S10 (*45*)]. Because this metric is normalized, the information that it provides remains independent of community size and diversity.

Our analysis reveals that the healthy state in the model is characterized by a negative net interaction, whereas the dysbiotic state is characterized by a positive net interaction (Fig. 3A), which we use to identify the two states in the model (*45*). This behavior is robust across different values of the parameters of the model (fig. S4). Moreover, fig. S12B shows that the duration of a state correlates with the strength of the interactions: More positive interactions are associated with longer-lasting dysbiotic states (similarly with negative interactions and healthy states). A more detailed view of these interactions reveals that dysbiotic states are characterized by a decrease in competition and an increase in cross-feeding (more specifically, commensalism and mutualism) compared with the healthy state (fig. S14).

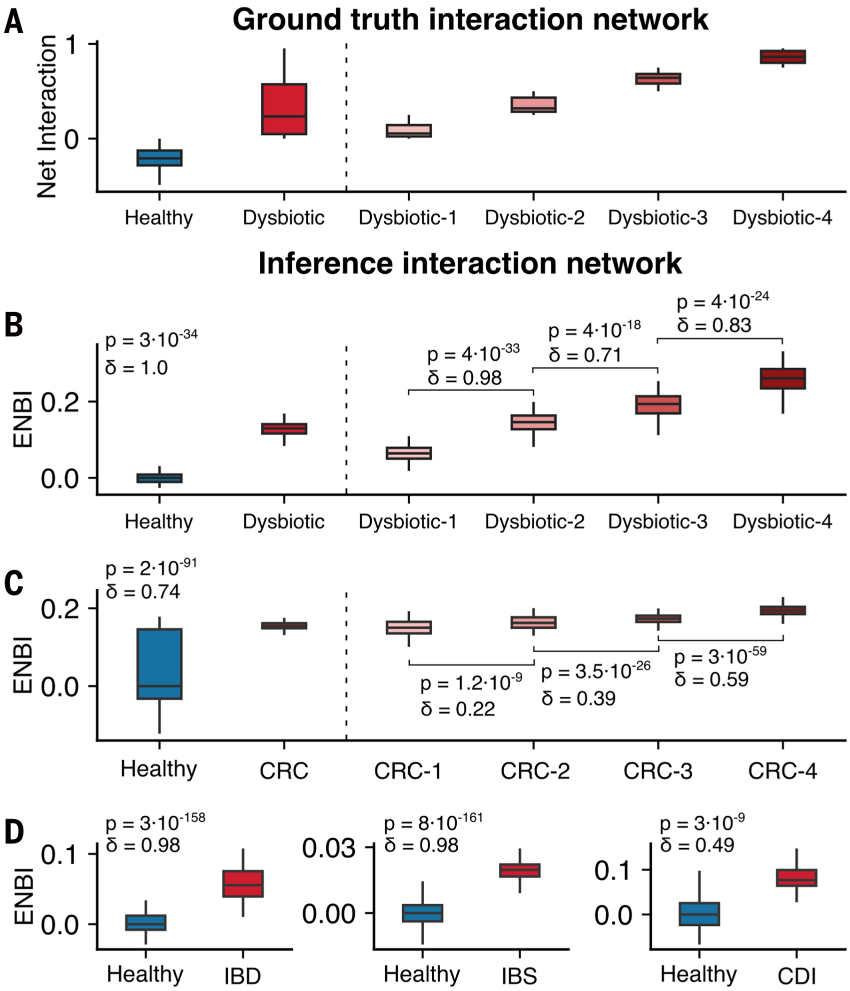

Fig. 3. The ENBI discerns healthy from pathological communities and correlates with disease progression. (**A**) Net interaction, ρ, calculated from the actual interaction network in the model for healthy and dysbiotic states separately (left side of the dashed line) and different consecutive configurations of unhealthy states (right side). For the latter, configurations were grouped and ordered in four sets (i.e., quartiles) from smaller to larger ρ. (**B**) ENBI calculated for model data, showing that this indicator captures the same qualitative behavior as the net interaction. (**C**) ENBI for healthy and CRC patients (left side of dashed line) showing that the latter have a positive ENBI, which suggests a more positively interacting microbial community compared with healthy patients; the ENBI increases with disease progression (right side), supporting our hypothesis of a progressive shift toward network structures with more positive interactions. (**D**) ENBI for IBD, IBS, and CDI also show that diseased states tend to show more positive interactions than their respective controls. See ENBI definition in eq. S12. Effect sizes (Cliff's delta) and *P* values (two-sided Mann-Whitney *U* test) were calculated as described in the materials and methods (*45*).

On the basis of these observations, we hypothesize that in IBD and potentially in other diseases, the gut microbiome shifts toward greater positive interactions compared with the dynamics observed in healthy controls. The net interaction, which focuses on the properties of the underlying interaction network, captures this transition. On the other hand, coarser metrics such as α diversity indices (e.g., the Shannon index) are broadly distributed for any given value of ρ (fig. S15), which may explain why diversity fails to consistently distinguish healthy from diseased states across conditions.

To test our hypothesis, we used an algorithm for empirical coabundance networks to infer effective interaction networks (*45*, *78*) and calculated the associated net interaction metric, ρ_{inferred}, for available real-world cross-sectional data and our model-generated data. The latter allowed us to validate the inferred interaction network against the actual one. The inferred net interaction ρ_{inferred} for different simulated communities captures the qualitative behavior of the actual underlying interaction network (Fig. 3, A and B): First, simulated communities from dysbiotic states show a higher ρ_{inferred} compared with those from healthy states; second, communities in dysbiotic states with higher true ρ values show higher values of ρ_{inferred}. Figure S16A shows that the inference method remains robust across different subsampling strategies. Building on this validation, we further applied this methodology across different diseases (IBD, CDI, IBS, and CRC). In all cases, there was a significant increase in ρ_{inferred} compared with their respective healthy control groups (fig. S16, B to E). The qualitative results remained consistent across additional network inference methods, including a method designed for cross-sectional data (*79*) (fig. S17) and another specifically developed for longitudinal data (*80*) (fig. S18).

We can use the healthy or control case as a baseline for each dataset and compute the difference between the ρ_{inferred} values of the dysbiotic and control inferred networks, which we called the ecological network balance index (ENBI) (eq. S12). This index is negative when the community exhibits dynamics with more negative interactions than the control (ENBI < 0) and positive when the community shows more positive interactions than the control (ENBI > 0).

Across all diseases analyzed as well as in our model, the dysbiotic state consistently exhibited a significantly positive ENBI value, reflecting a shift toward more positively interacting communities relative to the controls (Fig. 3, B to D). In the case of CRC, for which long-term data on disease progression are available, we further observed a progressive increase in ENBI as the disease advances, indicating a strengthening of positive interactions in the community over time (fig. S16B and Fig. 3D). Notably, this shift in ENBI is robust across a wide range of methodological and biological contexts. We observed consistent ENBI patterns across different geographic regions and, by extension, dietary backgrounds (fig. S19); across distinct taxonomic profiling pipelines (fig. S20); and across taxonomic resolutions, from the species to family level (fig. S21). Moreover, ENBI values remained similar for two independent healthy cohorts (fig. S22) as well as in a mild dietary intervention study involving a barley-rich diet versus a controlled diet (fig. S23). Together, these results support the robustness of the ENBI and reinforce its potential as a general indicator of disease-associated dysbiosis.

Discussion

Whether substantial changes in the gut microbiome cause or result from the onset of disease remains debated. In this work, we examined the ecological dynamics and community-level effects shaping the gut microbiome in health and disease. We developed the ENBI to quantify the dominant interaction dynamics, which offered insights into the structure of microbial communities and underlying mechanisms involved. Unlike traditional metrics (e.g., the Shannon index), the ENBI reveals a reproducible pattern across diseases and methodologies, highlighting a microbial shift toward more positively interacting communities in dysbiotic states compared with healthy controls.

Specifically, our model indicates that diseased states are associated with a shift in microbial interaction networks, by which overall positive interactions (commensalism and mutualism) increasingly dominate over negative interactions (competition). This relative increase reflects a global shift in the balance of microbial interactions within the community, which does not contradict previous studies describing reductions in specific cross-feeding interactions (*76*, *77*). Furthermore, in line with these previous studies, we also observe an increased metabolic self-sufficiency in dysbiosis: In the model, microbial communities in dysbiotic states are dominated by self-sufficient consortia composed of bacterial lineages that efficiently exploit available resources. These consortia can outcompete other species, potentially depleting key metabolites and altering or even eliminating functions important for host health. This exclusion often, but not always, leads to reduced α diversity, which explains why some diseases exhibit a loss of diversity whereas others do not.

Moreover, for colorectal cancer, for which data on disease progression are available, the ENBI correlates with disease advancement. If this trend holds across other diseases, the finding carries important clinical implications, particularly because our metric can be quantified noninvasively from stool metagenomic data. Therefore, monitoring the ENBI of a patient can help devise early warning indicators for disease onset and monitor progression, which is especially important for conditions such as CRC for which early detection is key to the success of the treatment and improvement of survival rates (*81*). Nonetheless, despite the broad applicability of the ENBI, the index by itself does not identify the specific disease causing dysbiosis. Thus, to maximize its diagnostic potential, the ENBI must be combined with other diagnostic tools for precise disease identification (*81*).

Our model follows a metabolically explicit approach, but, unlike genome-scale metabolic models and kinetic models (*39*), which require detailed genome annotations and well-defined objective functions with extensive parameter fitting, our simpler model focuses on essential ecological and metabolic principles and a biologically plausible parameterization. Our bottom-up approach thus has the potential to unveil the mechanistic underpinnings of gut microbiome emergent properties, such as diversity, multistability, and metabolic organization (*40*). For example, our model provides an explanation for the Anna Karenina principle (AKP)—the observation that the microbiomes of healthy individuals are seemingly similar to each other, but dysbiotic communities are all distinct (fig. S9). Healthy states in our model show similar functional composition, which constrains the communities to exhibit a certain degree of similarity, even though the taxa performing those functions may differ. By contrast, changes in the interaction network structure occurring in dysbiotic states partially or totally disrupt some of those functions in a nondeterministic manner that leads to more variable outcomes, with disease data pointing to potential disease-specific disruptions (*10*). Finally, the subtle behavior for β diversity observed in the model, and the lack of consistency observed in our analysis of β diversity across diseases, agree with the lack of generality of the AKP in empirical data, as proposed in the past (*82*).

Because the model captures ecological interactions in a broad and general way, it can be readily adaptable to other microbiomes, from other tissues (e.g., the vaginal or oral microbiomes) to plant or soil microbiomes. It can also easily expand to incorporate therapeutic interventions, such as probiotics or fecal transplantation. Nonetheless, there are limitations to be considered when using our framework. In the effort to isolate the core ecological mechanisms underlying the emergence of alternative community states, we deliberately excluded several processes that are known to influence gut microbiota composition. These include spatial structure, immune regulation, oxygen gradients, and explicit interactions with the host, currently represented in this work solely through resource input and dilution. Additionally, the model simplifies key aspects of microbial physiology; for example, it represents anabolic reactions indirectly (as opposed to explicitly) and uses a reduced parameterization for energy use and enzymatic costs. Other ecological factors, such as bacteriophage dynamics and chemical antagonism between bacteria (e.g., bacteriocin production), are also omitted. Expanding the model to incorporate a more realistic representation of these aspects is an important avenue for future work. Furthermore, our approach to analyzing empirical data relies on inferred ecological interactions, which intrinsically depend on data quality and quantity and come with inherent assumptions and challenges (e.g., a positive interaction may reflect either mutualism or coblooming owing to environmental change) (*38*, *83*). Other limitations are inherent to the heterogeneity of the methods used across studies. For example, cross-disease comparisons could provide valuable insights into microbiome dynamics for different conditions but are currently hindered by variability in sampling and different taxonomic profiling methods used across studies. Such methodological differences directly affect the inference of the ecological interaction network, which makes quantitative comparisons across diseases difficult. For that reason, we focused on identifying qualitative trends rather than absolute numerical differences. Addressing these challenges will require a multidisciplinary effort to integrate empirical studies and clinical data with theoretical approaches such as ours.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.ady1729
Materials and Methods; Figs. S1 to S24; Table S1; References (*97–112*);
MDAR Reproducibility Checklist

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Imbalance in gut microbial interactions as a marker of health and disease

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Editor's summary

A spectrum of diseases, from obesity to colorectal cancer, can arise from disrupted gut microbiota, which is known as dysbiosis. Dysbiosis is thought of as a shift from a healthy gut into an alternative steady state comprising different taxa with different metabolic functions. In a reexamination of empirical data, Corral López *et al*. made a consumer resource model to characterize bacterial and nutrient dynamics in the microbiota. The authors observed that simplified bacterial communities emerged in dysbiosis, and such communities efficiently metabolized the most energetic resources through fewer, more direct routes. The model showed that competitive interactions dominated in a healthy gut, whereas cooperative cross-feeding dominated in dysbiosis. How these ecological shifts are triggered is still to be understood, but observations of progression to greater cooperativity among the microbiota might be a sign of disease progression. —Caroline Ash

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